DS4002 – Case Study
Your Name:
Date:

Can You Outsmart Fake News?

In today's digital world, news spreads faster than ever, but not all of it is trustworthy. Social media and online platforms have made it increasingly difficult to distinguish real reporting from misinformation. Fake news now shapes public opinion across politics, health, economics, and global events, often using persuasive language to disguise false or misleading claims.

In response, media analysts have turned to data science to better understand how misinformation is written and spread. Rather than treating fake news detection as a single problem, researchers suggest it may depend on the topic, raising an important question: **are some types of news harder to classify than others?**

As a data scientist on a media analytics team, you are given a dataset of labeled news articles across multiple domains. Your task is to investigate how language patterns differ between real and fake news, and how these differences change across topics such as politics, science, entertainment, and world events.

By analyzing this data, you will help determine whether misinformation is uniformly detectable or whether certain topics allow fake content to blend in more effectively due to tone, vocabulary, or writing style.

The Deliverable:

In relevance to the aforementioned information, a media analytics team is seeking a data scientist to analyze a large dataset of labeled news articles in order to better understand how misinformation behaves across different topics. The goal is to determine whether patterns in language, structure, and vocabulary can be used to distinguish real from fake news, and how these patterns vary depending on subject matter.

Through your analysis, you are expected to uncover whether certain types of news are more susceptible to misinformation and to what extent the topic influences the detectability of fake news. Your work will contribute to a broader effort to improve how misinformation is understood and identified in digital media environments.

The results you provide may be used to inform the development of more effective detection strategies, helping researchers and analysts better understand how fake news adapts across different domains. In doing so, your findings should highlight which categories of news present the greatest challenges for accurate classification and what this reveals about the nature of misinformation in today's media landscape.